

MEMOREASON: EVALUATING THE EFFECT OF PARAMETRIC MEMORY ON CONTEXTUAL REASONING IN LLMs

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ABSTRACT

Large Language Models (LLMs) perform well on reasoning benchmarks, but it remains unclear whether this reflects genuine contextual reasoning or reliance on facts memorized in their parameters. We investigate this by distinguishing two possibilities: a broad *memory bias*, where familiar content improves reasoning performance, and the *Strong Parametric Shortcut Hypothesis*, where models skip reasoning entirely and recall stored answers. To test these effects, we introduce **MemoReason**, a human-curated benchmark that pairs factual reasoning tasks with structurally identical fictitious versions where real entities like people, companies, or dates are systematically replaced by fictitious ones of the same type. This keeps task complexity constant while varying the familiarity of the context, allowing controlled measurement of how the parametric memory affects reasoning. Our evaluation of recent LLMs reveals consistent and statistically significant performance drops of up to 15.7% in the fictitious setting, demonstrating a clear memory bias. However, a targeted analysis of questions failed in the fictitious setting shows that models rarely respond with the corresponding factual answer, indicating that direct parametric shortcuts are not the dominant failure mode. These findings suggest that parametric memory influences reasoning through mechanisms more complex than simple factual recall. **MemoReason** provides a controlled framework for studying these mechanisms and for extending paired factual–fictitious evaluation to broader reasoning settings¹.

1 INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkable performance in complex reasoning tasks by scoring highly on benchmarks involving math (Hendrycks et al., 2021; Lightman et al., 2023; Alshammari et al., 2026), code generation (Chen et al., 2021; Jimenez et al., 2024; Jain et al., 2024; Zhuo et al., 2025; Ye et al., 2025), commonsense inference (Zellers et al., 2019; Talmor et al., 2019), multi-hop question answering (Yang et al., 2018) and professional domain exams such as law and medicine (Katz et al., 2024; Nori et al., 2023). However, the extent to which this performance is due to abstract reasoning, pattern matching or factual memorization remains an open question. In this work, we propose an approach to better elucidate this question by studying LLM performance on reasoning tasks under controlled contextual information environments.

Memory Bias in Reasoning & the Strong Parametric Shortcut Hypothesis. Some studies show that, when performing reasoning tasks, LLMs can over-rely on their parametric memory – i.e. the knowledge stored in the model parameters during training – rather than on contextual knowledge (Longpre et al., 2021; Wu et al., 2024b). In this work, we begin by testing whether familiarity with the contextual information being processed biases the ability of a model to perform reasoning tasks. This relates to the question: *what is the impact of parametric memory on the reasoning performance of LLMs?* If memory plays an important role, reasoning performance should degrade when the model is exposed to new information that has similar textual patterns to the training data but fundamentally

¹Dataset: <https://huggingface.co/datasets/memoreason-anonymous/MemoReason>
Code: <https://anonymous.4open.science/r/MemoReason-D40A/>

different meaning. Henceforth we will refer to such information as *fictitious* (as opposed to *factual* information).

Additionally, we explore the *Strong Parametric Shortcut Hypothesis*: the idea that LLMs leverage embedded memory as an easy way out of reasoning tasks. In short, the hypothesis suggests that, rather than performing all the intermediary reasoning steps logically necessary to arrive at the final answer (e.g., arithmetic, relational inference, temporal reasoning), models proceed by wrongly recalling a memorized response.

For mathematical reasoning, works such as `GSM-Symbolic` (Mirzadeh et al., 2025) and `MATH()` (Srivastava et al., 2024) have successfully isolated true reasoning from memorization (due to benchmark leakage into training data) by replacing entities within fixed problem templates, thereby maintaining the same task structure and complexity.

For reasoning tasks over semantically-rich documents, recent literature has sought to investigate shortcuts (Glockner et al., 2025) and *memory biases* (Wu et al., 2024a). In the case of the `CoFca` benchmark by Wu et al. (2024a), a difference in reasoning performance between factual and fictitious examples was reported. However, the gap they reported could be fully explained by the unaccounted systematic error introduced by the fact they used two **distinct** sets of evidence-question pairs for the factual and the fictitious data. Hence, a strong conclusion on the role of parametric memory in reasoning cannot be drawn from their work.

In this paper, we set out to close this gap by applying the robust approach by Mirzadeh et al. (2025) to semantically-rich documents. We do so by introducing **MemoReason (Memorization in Reasoning)**, a novel, high quality human-curated benchmark designed specifically to test *memory biases* in reasoning and the *Strong Parametric Shortcut Hypothesis* under strictly controlled conditions.

We evaluate LLMs on reasoning tasks using known factual entities and subsequently test them on the exact same tasks where those familiar entities are replaced with fictitious ones under rule constraints. To the best of our knowledge, we are the first to propose a benchmark that compares LLM performance on a complex, human-curated, multi-domain question-answering dataset under these controlled conditions. We make the following main contributions:

- We introduce **MemoReason**, a novel reasoning question-answering benchmark that pairs factual tasks with structurally identical fictitious counterparts.
- We formulate a rigorous framework to evaluate how much LLMs rely on contextual reasoning and parametric shortcuts using this novel benchmark.
- We open-source the customized annotation interface used to build **MemoReason** as a commitment to help future work aiming to extend our approach.
- We show that LLMs exhibit significant performance drops on reasoning questions when moving from factual to fictitious settings.
- We verify that direct factual information recall (the *Strong Parametric Shortcut Hypothesis*) is not the dominant memory bias in recent LLMs

2 BENCHMARK CONSTRUCTION

To examine familiarity biases in reasoning, we define a setup that allows us to systematically isolate the parametric memory without changing the underlying reasoning. To this end, we start by consolidating a factual dataset based on Wikipedia excerpts (Section 2.1). We then use this dataset to build templates that preserve the text structure while abstracting its content (Section 2.2). Finally, we generate fictitious variants of the factual dataset (Section 2.3).

2.1 FACTUAL DATASET

We start by building a factual dataset $\mathcal{D} = \{(d_i, (q_{i,j}, a_{i,j})_{j=1}^Q)\}_{i=1}^N$ where d_i is an excerpt of a Wikipedia article (introduction section), and $(q_{i,j}, a_{i,j})$ a question-answer tuple based on d_i . Like `GSM-Symbolic`, which instantiates 100 templates, **MemoReason** is organized around $N=100$ source templates; here each template is an entity-rich document supporting structured interventions

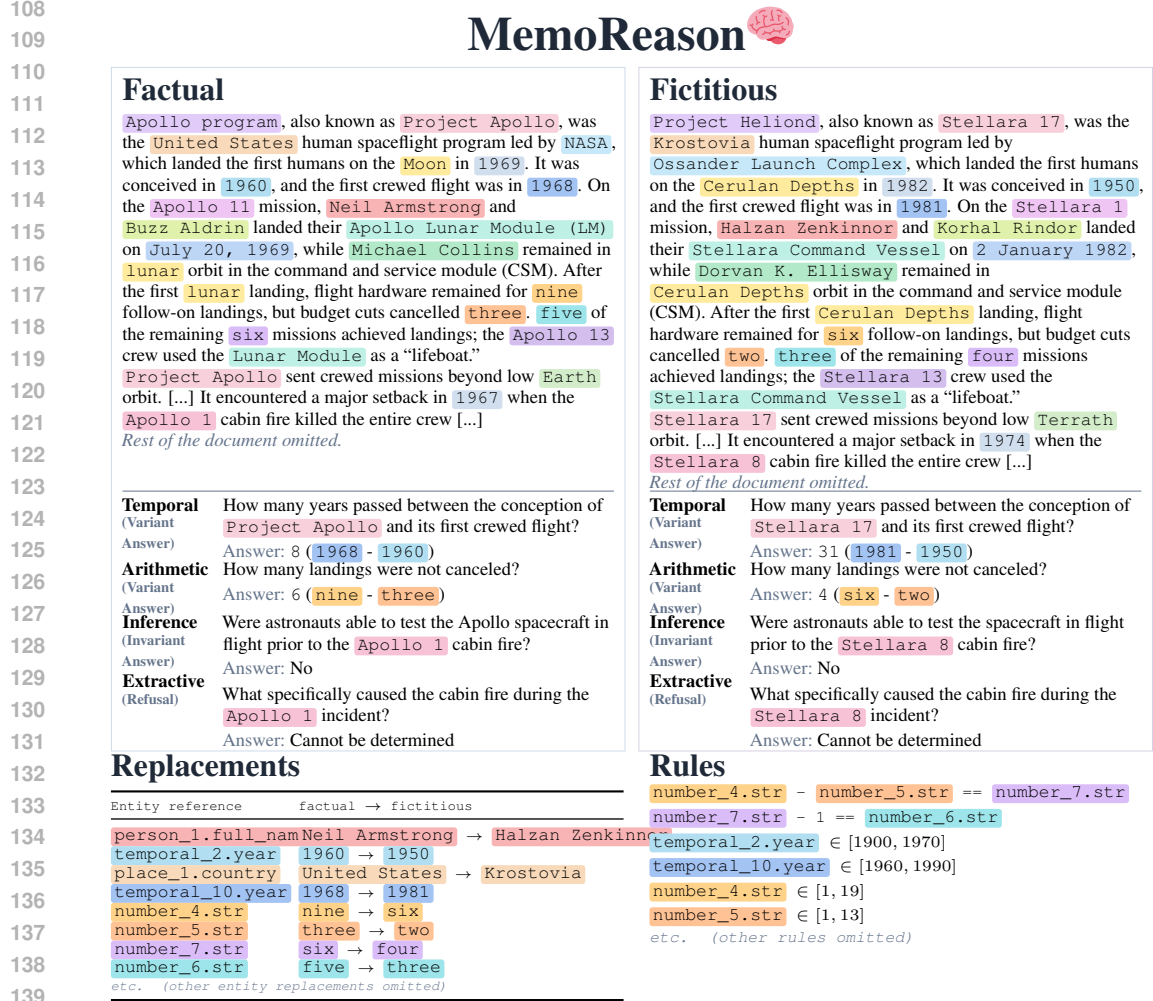


Figure 1: Illustration of the MemoReason benchmark on the Apollo Program template showcasing the factual excerpt from Wikipedia on the top left with its annotated entities, a paired fictitious document on the top right, the factual and fictitious question-answer pairs below, the replacement table that maps each factual entity to its fictitious value and some of the associated rules that should be satisfied when generating the fictitious variants on the bottom. The same highlighting color is used to represent two paired entities between factual and fictitious.

and paired variants. Each document template has $Q = 12$ question-answer (Q-A) pairs, for a total of 1,200 factual task templates; Table 1 also reports their character lengths.

We chose Wikipedia excerpts treating famous topics, as it is reasonable to assume that most LLMs were exposed to them during training. The selected documents contain a high density of entities, making them ideal for probing the parametric memory of LLMs. Finally, the dataset spans nine diverse themes for which the statistics are reported in Table 1 (additional details available in Appendix E.1).

The question-answer pairs associated with each document are chosen to cover four *question classes* and three *answer classes* (see Section 2.3).

Question Classes. Four classes of questions are proposed: simple extractive questions, arithmetic questions involving calculations, inference questions that are not numerical, and temporal reasoning. We propose these question classes to test *memory biases* on different reasoning tasks (see Figure 1 for examples of each):

Table 1: **Dataset statistics by theme category.** Document and task-template counts are reported by theme; entity and rule counts are per document ($\pm$ sample standard deviation).

	<i>Award Winners</i>	<i>Biographies</i>	<i>Places</i>	<i>Companies</i>	<i>Natural Disasters</i>	<i>Public Attacks</i>	<i>Retail Banking</i>	<i>Space Missions</i>	<i>Sport Events</i>	<i>Total</i>
#Documents	12	10	10	12	12	10	11	12	11	100
#Task templates	144 (12.0%)	120 (10.0%)	120 (10.0%)	144 (12.0%)	144 (12.0%)	120 (10.0%)	132 (11.0%)	144 (12.0%)	132 (11.0%)	1,200
Avg length (chars)	2,974 $\pm 1,036$	3,160 ± 952	2,606 ± 600	1,943 ± 799	2,644 ± 969	2,454 $\pm 1,337$	1,265 ± 498	1,503 ± 698	2,268 ± 750	2,298 $\pm 1,037$
Avg #Entities	58.0 ± 26.1	51.4 ± 14.6	48.7 ± 14.4	41.4 ± 14.9	39.8 ± 14.2	22.3 ± 9.5	14.3 ± 5.5	26.2 ± 9.8	39.4 ± 12.6	38.0 ± 19.6
Avg #Rules	57.2 ± 34.7	35.5 ± 9.6	31.4 ± 8.8	21.7 ± 13.0	38.5 ± 16.4	17.4 ± 9.6	8.4 ± 5.6	23.5 ± 12.8	44.9 ± 15.7	31.2 ± 21.4

- **Extractive (baseline):** the answer is explicitly stated in the document, so the main challenge is to identify and extract the relevant evidence.
- **Arithmetic:** answering this type of question requires performing arithmetic operations on the numerical entities mentioned in the document.
- **Temporal:** this type of question aims to test the ability of models to reason over temporal entities such as years, dates, ages, or durations.
- **Inference:** answering this type of question requires reasoning that is not primarily arithmetic or temporal.

2.2 TEMPLATE ANNOTATION

The template annotation process consists of transforming the factual dataset $\mathcal{D}$ into a dataset of templates $\mathcal{D}^{\text{template}} = \{(d_i^{\text{template}}, (q_{i,j}^{\text{template}}, a_{i,j}^{\text{template}})_{j=1}^Q, \mathcal{E}_i^{\text{factual}}, \mathcal{R}_i)\}_{i=1}^N$ where d_i^{template} is a document with annotated entities following a taxonomy that we define in Appendix B, $q_{i,j}^{\text{template}}$ the question, $a_{i,j}^{\text{template}}$ the answer, $\mathcal{E}_i^{\text{factual}}$ the set of factual entities, and $\mathcal{R}_i$ the set of rules that must be satisfied when replacing factual entities with fictitious ones in d_i^{template} . Unlike GSM-Symbolic (Mirzadeh et al., 2025), which operates on short arithmetic statements, our setting involves entity-rich documents with substantially greater complexity, requiring us to design more sophisticated and expressive templates.

Entity Annotation. A factual document d_i is transformed into a template document d_i^{template} by identifying entities and assigning the right entity type and attribute (e.g., *Albert Einstein* gets assigned the person entity type with a `full_name` attribute). This annotation process ensures that factual entities are replaced with plausible fictitious ones that preserve their type (e.g., person $\rightarrow$ person, number $\rightarrow$ number). To this end, we developed a taxonomy of 14 entity types and their associated attributes, designed to cover all entities present in the factual dataset (see Appendix B for details).

Rules. It is necessary to make sure that replacing the factual entities does not alter the logic and core semantics of the document. We therefore manually encode constraints or rules that must be satisfied when replacing with fictitious entities. Those rules include:

- Plural mentions: "[...] *He was among the [10; number_1.int] selected players* [...]"
 $\rightarrow \text{number_1.int} \geq 2$
- Numbers that should sum to a total: "[...] *Among the [fourteen; number_2.str] casualties are [9; number_3.int] injured and [5; number_4.int] dead* [...]"
 $\rightarrow \text{number_2.str} = \text{number_3.int} + \text{number_4.int}$

- Order preservation: we define rules to ensure that the order of numerical and temporal entities (e.g. years, dates, etc.) is maintained between factual and fictitious settings. This is crucial especially for temporal entities to avoid breaking the logical timeline of events.
- Sampling intervals: for number and temporal entities, we sample the fictitious replacements from restricted intervals around the factual values to avoid situations where replacing a small number with a very large one would create unnatural situations such as asserting a historical figure lived to be 800 years old instead of 80 or that a stock price went negative.

Annotation Process. All the annotation steps, including the entity annotation, rule definition, and question-answer writing are first drafted by an AI Agent backed by Claude Opus 4.6 (Anthropic, 2026a). Then human annotators (qualified colleagues with knowledge in LLMs) independently check the generated annotations and modify them if needed. Finally, the annotators meet in a dedicated agreement session. Even with a highly capable AI Agent, systematic human annotations proved indispensable for correcting, refining and validating all initial AI drafts. As a point of reference, almost 50% of the Q-A pairs required human intervention. We stress that the annotation and review process is both intensive and of critical importance. We show in Figure 2 a screenshot of the annotation interface used to inspect entity spans, replacement rules, and question-answer fields for the Toyota template. Although Claude Opus 4.6 drafts fictitious named entities and Claude Sonnet 4.6 is among the evaluated models, candidate acceptance is model-independent: entities undergo web and Wikipedia non-existence checks, human review, and deterministic rule validation before evaluation.

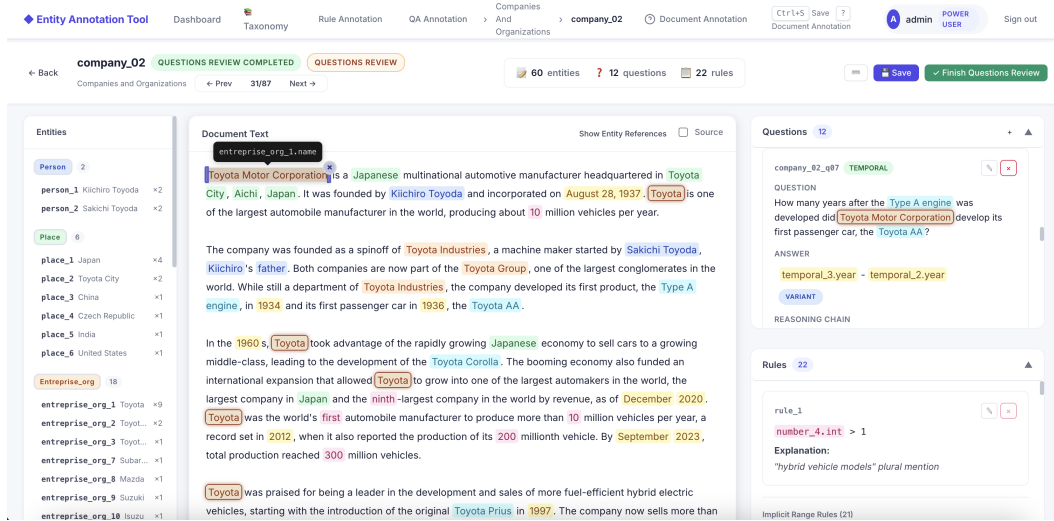


Figure 2: Screenshot of the annotation interface used to inspect entity spans, replacement rules, and question-answer fields for the Toyota template.

2.3 FICTITIOUS DATASET

For each template, we generate $K = 10$ fictitious variants $\{(d_{i,s}^{\text{fictitious}}, (q_{i,j,s}^{\text{fictitious}}, a_{i,j,s}^{\text{fictitious}})_{j=1}^Q)\}_{s=1}^K$ by replacing all entities under rule constraints. We categorize the entities into **named** entities (e.g. persons, places, etc.) and **numerical** entities (e.g. numbers, temporal entities such as dates, etc.). To ensure that the generated entity values are fictitious and satisfy the replacement rules, the generation process includes 2 steps for each document:

- **Named entity sampling.** We use an AI Agent backed by Claude Opus 4.6 (Anthropic, 2026a) to build a document-specific pool of fictitious named-entity candidates. The generated candidates must be non-existing, pronounceable (BAUER, 2015), type-compatible, and internally coherent across attributes. When a document links several attributes of a single entity, such as a place name and its demonym, the candidate pool keeps the linked attributes aligned by variant index so that a sampled fictitious document remains coherent.

The prompt used is provided in Appendix J.5. We also ensure that the generated entities do not exist by checking their existence in Wikipedia and via a web search.

- **Numerical entity sampling.** To sample numerical and temporal entities that satisfy the rule constraints, we use the Mixed Integer Linear Programming solver implemented in `SciPy` (Virtanen et al., 2020). We also make sure not to sample the same values across the K fictitious variants. Days and months are replaced by Monte Carlo sampling and verified manually against the rules.

Answer Classes. Orthogonally to question class, and in order to systematically isolate the influence of parametric memory, we define three answer classes. Questions with *variant* answers depend on the entities described in the document and, therefore, their answers change when those entities are replaced. Questions with *invariant* answers are independent of the fictitious substitutions and remain identical between factual and fictitious settings. Invariant questions are still evaluated on the fully fictitious document—both named identities and numerical/temporal values are replaced—but their answer literal is unchanged. *Refusal* questions ask for information that is not provided in the document, testing whether the model stays grounded in the provided context.

Each template is used to generate $K = 10$ fictitious variants for each of its $Q = 12$ Q-A pairs, yielding 12,000 fictitious pairs. Figure 1 illustrates a template on the Apollo Program example.

3 EVALUATION FRAMEWORK

Measuring Performance Variation. Given the performance of a model on a factual example $(d_i, q_{i,j}, a_{i,j})$, noted $Y_{i,j}^{\text{factual}}$, we are interested in how this performance changes towards the set of K corresponding fictitious examples $\{(d_{i,s}^{\text{fictitious}}, q_{i,j,s}^{\text{fictitious}}, a_{i,j,s}^{\text{fictitious}})\}_{s=1}^K$. We define the paired difference as $\Delta_{i,j} = \hat{Y}_{i,j}^{\text{fictitious}} - Y_{i,j}^{\text{factual}}$, where $\hat{Y}_{i,j}^{\text{fictitious}}$ denotes the average performance across the K corresponding fictitious examples. To account for dependence among questions and variants from the same document, confidence intervals are computed by bootstrapping complete documents.

Exact and Judge Match. We assess prediction correctness using two complementary metrics. We first apply Exact Match (EM) and for predictions deemed incorrect we follow up with Judge Match (JM), in which an LLM is prompted to determine whether the prediction and the ground truth are semantically equivalent. To validate the reliability of the JM, we calibrated it on 200 randomly sampled examples, achieving 100% human-machine agreement. This confirms that the JM not only captures correct predictions missed by EM, but also reliably identifies incorrect ones, suggesting it is not prone to the confirmation bias toward affirmative judgments that has been reported in LLMs (Jain et al., 2025; He et al., 2026). We provide more details on the calibration process in Appendix C.

Model Selection. We evaluate nine recent, high-performing reasoning models spanning multiple providers: GPT-OSS-20B and GPT-OSS-120B (OpenAI et al., 2025), OLMo-3-7B-Think and OLMo-3-7B-Instruct (Olmo et al., 2026), Gemma-4-26B-A4B-it, Qwen3.5-27B and Qwen3.5-35B-A3B (Qwen, 2026), Llama-3.1-8B-Instruct (Dubey et al., 2024), and Claude Sonnet 4.6 (Anthropic, 2026b;a). This set combines open-weight models with a frontier API model, allowing us to compare memory biases and parametric-shortcuts across model families and scales.

4 EXPERIMENTS & RESULTS

In this section, we describe the experiments conducted on the **MemoReason** benchmark. In Section 4.1 we examine the impact of replacing entities in the document on LLMs’ ability to answer questions based on contextual information, testing the existence of a *memory bias* in reasoning. In Section 4.2, we study how performance evolves as the proportion of replaced entities increases, testing whether a mixture of real and fictitious entities creates additional conflict between parametric and contextual knowledge. Section 4.3 tests chain-of-thought effects and reasoning effort. Finally, in Section 4.4, we put the *Strong Parametric Shortcut Hypothesis* to the test to ascertain whether prediction errors are due to pure recall of learned facts. Additional robustness and mechanism checks are reported in Appendix G.1.

Table 2: Mean performance change (fictional minus factual, percentage points) by answer and question type. Brackets are 95% CIs; significant cells are framed and bold. *Reason.* aggregates arithmetic, temporal, and inference questions. Purple and blue denote decreases and increases, respectively.

Question	Variant					Invariant					Refusal				
	Arith.	Temp.	Infer.	Reason.	Extr.	Arith.	Temp.	Infer.	Reason.	Extr.	Arith.	Temp.	Infer.	Reason.	Extr.
OLMO-3 7B-THINK	-1.4 [-9.2, +6.5]	-15.6 [-23.2, -7.8]	-7.4 [-14.6, +0.0]	-8.1 [-12.4, -3.8]	-9.3 [-13.6, -4.8]	-0.8 [-8.3, +6.7]	-6.6 [-13.2, +0.4]	-2.2 [-10.2, +6.3]	-3.2 [-7.7, +1.4]	-1.6 [-7.7, +4.6]	-3.6 [-9.1, +2.0]	-10.8 [-17.9, -3.6]	-2.0 [-7.9, +4.0]	-5.5 [-9.1, -1.9]	-2.4 [-7.1, +2.5]
OLMO-3 7B-INSTRUCT	-4.0 [-12.9, +4.6]	-3.7 [-11.5, +4.1]	-15.7 [-24.8, -6.5]	-7.8 [-12.6, -2.9]	-6.0 [-12.0, +0.1]	-1.6 [-8.0, +4.7]	-11.1 [-18.5, -4.1]	-5.4 [-12.6, +1.7]	-6.0 [-10.1, -2.1]	-0.3 [-5.7, +5.2]	+2.1 [-2.8, +7.4]	+6.5 [+0.0, +13.2]	+3.9 [-1.7, +9.9]	+4.2 [+0.8, +7.6]	+1.4 [-2.1, +5.3]
GPT-OSS 20B	-4.8 [-9.3, -0.5]	-4.8 [-9.3, -0.6]	-11.3 [-18.2, -4.5]	-7.0 [-10.2, -3.8]	-7.0 [-12.0, -2.6]	-0.3 [-6.6, +6.2]	-9.7 [-15.6, -4.1]	+0.2 [-4.2, +4.8]	-3.3 [-6.5, -0.1]	-2.2 [-6.4, +1.8]	-3.6 [-8.6, +1.1]	+1.8 [-3.7, +7.4]	+4.1 [-2.6, +11.0]	+0.8 [-2.7, +4.3]	+0.4 [-3.5, +4.5]
GEMMA-4 26B-A4B-IT	-6.9 [-13.2, -0.7]	-4.6 [-10.3, +1.0]	-8.1 [-15.1, -1.4]	-6.5 [-10.2, -2.9]	-4.2 [-7.7, -1.4]	+0.1 [-5.8, +6.0]	-2.0 [-7.2, +3.3]	+0.5 [-5.0, +6.1]	-0.5 [-3.7, +2.7]	-0.0 [-3.3, +3.3]	+3.8 [+0.7, +7.8]	+3.4 [-1.0, +8.2]	-0.2 [-4.5, +4.2]	+2.3 [+0.1, +4.7]	+2.1 [-0.6, +5.6]
GPT-OSS 120B	-4.7 [-9.3, -0.2]	-7.4 [-11.3, -4.0]	-11.8 [-18.8, -4.9]	-8.0 [-11.3, -4.7]	-7.8 [-12.5, -3.7]	-0.8 [-6.6, +5.2]	+2.1 [-3.2, +7.6]	-0.2 [-3.3, +3.3]	+0.4 [-2.6, +3.3]	-0.6 [-4.4, +3.2]	-5.6 [-10.7, -0.8]	-1.0 [-5.0, +2.9]	-1.4 [-6.5, +3.7]	-2.7 [-5.4, +0.1]	-1.4 [-5.5, +2.7]
QWEN3.5 27B	-5.3 [-10.2, -0.8]	-4.4 [-9.2, -0.0]	-10.0 [-17.7, -2.5]	-6.6 [-10.1, -3.1]	-5.0 [-9.0, -1.7]	+0.6 [-5.5, +6.9]	-3.4 [-7.4, +0.5]	-2.1 [-5.5, +0.7]	-1.6 [-4.6, +1.3]	-3.1 [-6.5, -0.5]	-4.6 [-9.8, +0.2]	-1.0 [-5.3, +3.0]	+2.5 [-2.0, +7.4]	-1.0 [-3.8, +1.8]	+0.4 [-1.3, +2.9]
QWEN3.5 35B-A3B	-6.5 [-11.8, -1.5]	-2.6 [-8.3, +2.9]	-10.4 [-16.1, -5.0]	-6.5 [-9.9, -3.2]	-5.4 [-10.3, -0.9]	-3.7 [-10.2, +2.5]	-3.7 [-9.4, +1.9]	-5.1 [-10.1, -0.7]	-4.2 [-7.5, -1.0]	-3.3 [-7.0, -0.2]	+2.2 [-2.4, +7.2]	+1.0 [-2.1, +4.5]	+2.0 [-0.3, +5.2]	+1.7 [-5.4, +0.1]	+2.6 [-0.2, +6.2]
LLAMA-3.1 8B-INSTRUCT	-3.0 [-9.5, +3.6]	-4.1 [-9.7, +1.4]	-12.6 [-19.1, -6.4]	-6.6 [-10.4, -2.8]	-1.9 [-6.2, +2.4]	+2.5 [-3.0, +8.2]	+2.3 [-4.9, +9.6]	-0.3 [-2.7, +2.1]	+1.5 [-1.6, +4.6]	-1.1 [-5.4, +2.9]	-1.9 [-6.8, +2.8]	-2.3 [-9.6, +5.0]	+3.1 [-2.8, +9.4]	-0.4 [-4.2, +3.5]	+2.9 [-3.5, +9.5]
CLAUDE SONNET 4.6	-7.2 [-12.4, -2.1]	-6.7 [-10.8, -3.1]	-9.6 [-15.1, -4.6]	-7.8 [-10.8, -5.0]	-3.1 [-7.3, +0.8]	-1.5 [-7.3, +4.4]	-0.4 [-4.1, +3.3]	-3.2 [-7.0, +0.4]	-1.7 [-4.5, +1.0]	-1.7 [-5.4, +1.8]	-5.3 [-9.5, -1.7]	+1.9 [-2.7, +6.8]	-2.4 [-7.0, +2.1]	-1.9 [-4.5, +0.6]	-1.2 [-4.3, +1.7]

4.1 THE IMPACT OF ENTITY REPLACEMENT ON LLMs' REASONING PERFORMANCE

Our first experiment investigates whether entity replacement influences model reasoning performance. The results are reported in Table 2. Per-class results are primary because each document contributes one question to each answer-question cell; the pooled *Reason.* column is retained as a summary (i.e. aggregation across arithmetic, temporal, and inference questions), with its interval computed under the same document-level resampling.

Reasoning questions with Variant answers For variant answers (see Section 2.3), all models exhibit a drop in reasoning performance when working on fictitious documents. Moreover, this drop is statistically significant for all 9 models tested. This result is driven by inference-class questions (see Section 2.1), with significant drops for 8 models in the 8.1% to 15.7% range. Temporal-class questions also contribute, with significant drops for 5 models in the 4.4% to 15.6% range. Arithmetic-class questions show significant drops for 6 models, in the 4.7% to 7.2% range. This ranking should not come as a surprise, since arithmetic questions are much more prevalent than other categories of questions on many mathematical reasoning benchmarks and post-training datasets. Temporal questions, though much less prevalent, are relatively straightforward to systematize and generally share similar patterns among each other. Inference questions, however, are much broader in the patterns they follow: they are the most difficult ones to shortcut via pattern matching. Overall, the Variant-class results confirm that there is indeed a memory bias in LLM reasoning.

Reasoning questions with Invariant answers For invariant answers, aggregate changes are smaller and more heterogeneous, ranging from a 6.0-point decrease to a 1.5-point increase. Only 3 of the 9 models show a significant aggregate decrease, and the per-class effects do not follow a uniform pattern across models. Appendix G.1.1 further shows that the relevant named-versus-numeric cues are model-dependent.

Refusal answers and Extractive questions. The picture is significantly more muddled for refusals, with aggregate changes ranging from a 5.5-point decrease to a 4.2-point increase and 3 significant changes going in different directions. Overall, a non-statistically significant hint of an inverse bias can be gleaned from the data, with models being more likely to venture an answer despite insufficient supporting evidence when in the presence of factual information. This familiarity-driven overconfidence remains a hypothesis rather than a general finding. On extractive questions with variant answers, performance decreases for all 9 models and significantly for 6. The *Strong Parametric Shortcut Hypothesis* still cannot be resolved from Table 2 alone; we address it in Section 4.4.

4.2 VARYING THE PROPORTION OF REPLACED ENTITIES

Section 4.1 compares factual and fully fictitious documents. Here, we vary the exact replacement rate from 0% to 100%, preserving every template rule, to probe knowledge conflict when factual anchors and counterfactual relations coexist (see Figure 3 for Qwen3.5-27B and Figure 6 in Appendix G.2 for the rest of the models).

All seven curves fall through 50% replacement: as familiar entities disappear, so does their advantage. Six then recover after minima between 50% and 80%; OLMo-3-7B-Instruct changes by only 0.07 points and plateaus. The rebound suggests an added cost when familiar entities conflict with counterfactual relations in the document, a conflict absent at the fully fictitious endpoint (Longpre et al., 2021). Qwen3.5-27B, for example, falls from 91.75% to 88.09% at 80%, then recovers to 88.80%.

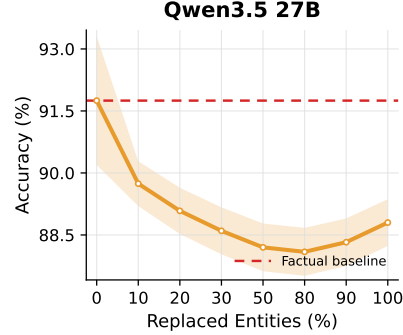


Figure 3: Qwen3.5-27B partial-replacement curve.

4.3 CHAIN OF THOUGHT EFFECT

Chain-of-thought comparison. CoT prompts intermediate steps intended to improve contextual reasoning. We test whether it also improves entity-replacement robustness. On reasoning questions, OLMo Think (CoT trace) gains 15.00/12.62 factual/fictitious points over Instruct (no trace), but its gap is larger (5.60 vs. 3.22): accuracy rises, robustness does not. We show a comparison example in Appendix H.1.

Controlled reasoning effort. Unlike GPT-OSS-20B, OLMo Think was not trained to support selectable reasoning-effort levels and therefore provides no inference-time control over its reasoning budget. We thus use GPT-OSS-20B to ask whether more effort helps within one model, varying low, medium, and high via `reasoning_effort`; higher settings elicit longer CoT on average (OpenAI et al., 2025; Olmo et al., 2026). Factual/fictitious accuracy rises 3.50/2.25 points low-to-medium but falls 1.58/1.59 at high (paired 95% CIs exclude zero), so more effort is not always better. The gap is significant throughout but its pairwise changes are not; more effort does not remove it.

Table 3: GPT-OSS-20B effort ablation.

Effort	Factual acc.	Fictitious acc.	Gap (pp)
Low	90.75	87.65	3.10
	[89.17, 92.25]	[86.15, 89.08]	[1.44, 4.77]
	94.25	89.90	4.35
Medium	[93.00, 95.50]	[88.35, 91.40]	[2.89, 5.86]
	92.67	88.31	4.36
High	[91.17, 94.17]	[86.68, 89.87]	[2.80, 5.98]

4.4 STRONG PARAMETRIC SHORTCUT HYPOTHESIS

The performance drops observed in fictitious variants (see Section 4.1) show that models are affected by the familiarity of the entities appearing in the document. However, this result alone does not establish the *Strong Parametric Shortcut Hypothesis*. A drop in performance may indicate a broader memory or familiarity bias, but the hypothesis would require a more specific failure mode: when the model fails on a fictitious variant, it should incorrectly answer with the factual answer associated with the original example, suggesting that it bypassed contextual reasoning completely by recalling memorized parametric knowledge.

To test this hypothesis directly, we compute the *Shortcut Rate* reported in Table 4. This measures how often the model’s incorrect prediction matches the corresponding factual answer. For each model, we restrict the analysis to failed fictitious variant examples. The resulting rates are consistently low. Across reasoning questions, the aggregate rate ranges from 1.0% to 3.2%, and extractive rates range from 0.0% to 1.0%. Thus, even when models fail under fictitious substitutions, they rarely fail by simply reproducing the factual answer.

These results suggest that the memory effect identified in our main performance analysis should not be interpreted as a simple recall shortcut. Parametric memory appears to influence reasoning performance, producing a clear *memory* or *familiarity bias*, but this influence does not usually translate into direct factual-answer copying. Instead, the failures likely reflect a more complex interaction between parametric knowledge and contextual reasoning, where familiar entities may affect retrieval priors, confidence, attention to constraints, or the stability of intermediate reasoning steps without fully determining the final response. The non-trivial interaction between prior knowledge and reasoning ability deserves further investigation, and **MemoReason** provides a controlled setting for analyses. Appendix G.1 reports directional flips, cue ablations, the reverse intervention, and robustness checks over temperature and training-data exposure.

4.5 BENCHMARK EVALUATION CONTRACT

MemoReason is intended not only to detect familiarity effects, but also to support meaningful comparisons between models. A single score would be misleading: fictitious accuracy reflects overall reasoning ability, whereas a small factual–fictitious gap may indicate robustness or simply poor performance in both settings. We therefore report complementary diagnostics. Table 5 ranks models by fictitious accuracy, while also reporting the factual–fictitious gap, the error ratio—which contextualizes this gap relative to the model’s remaining errors—and the Shortcut Rate, which measures direct factual-answer reproduction among fictitious failures. These metrics should be interpreted jointly. For example, Llama-3.1-8B-Instruct has a relatively small 1.37-point gap but ranks seventh in fictitious accuracy, and its reasoning performance still decreases significantly.

5 CONCLUSION

We introduced **MemoReason**, a human-curated benchmark for studying how parametric memory affects contextual reasoning in LLMs. By pairing factual tasks with structurally identical fictitious counterparts, **MemoReason** keeps the reasoning problem fixed while changing the familiarity of the entities in the context. Our evaluation of recent LLMs reveals consistent and statistically significant performance drops of up to 15.7% in the fictitious setting, demonstrating a clear memory bias. However, our analysis shows that these failures are rarely direct factual-answer recalls. Thus, parametric memory affects reasoning performance, but not through a simple shortcut in which the model copies the factual answer. Instead, our results point to a more complex interaction between memorized knowledge and contextual reasoning, where entity familiarity may influence retrieval priors, confidence, or intermediate reasoning without directly determining the final response.

Table 4: Shortcut Rate (%). Rate = judge-weighted matches over failed variant examples (counts in parentheses). *Reason.* aggregates reasoning types.

Question Type	Arith.	Temp.	Infer.	Reason.	Extr.
OLMO-3 7B-THINK	1.0 (344)	1.4 (336)	0.5 (314)	1.0 (994)	0.8 (123)
OLMO-3 7B-INSTRUCT	2.0 (600)	3.7 (567)	1.9 (467)	2.5 (1634)	0.8 (130)
GPT-OSS 20B	1.6 (118)	3.9 (98)	0.9 (213)	2.2 (429)	1.0 (80)
GEMMA-4 26B-A4B-IT	1.4 (189)	0.6 (186)	1.3 (271)	1.1 (646)	1.0 (52)
GPT-OSS 120B	0.8 (107)	2.0 (94)	1.2 (188)	1.3 (389)	1.0 (78)
QWEN3.5 27B	1.9 (253)	0.4 (134)	1.2 (200)	1.2 (587)	1.0 (60)
QWEN3.5 35B-A3B	1.6 (305)	1.6 (186)	3.9 (254)	2.3 (745)	1.0 (74)
LLAMA-3.1 8B-INSTRUCT	1.8 (360)	2.6 (271)	5.2 (256)	3.2 (887)	0.0 (39)
CLAUDE SONNET 4.6	1.2 (102)	1.4 (87)	3.3 (136)	2.0 (325)	1.0 (41)

Table 5: 🏆 **MemoReason leaderboard.**

Rank	Model	Fictional acc. (%)	Gap (%)	Error ratio	Shortcut Rate (%)
1	CLAUDE SONNET 4.6	92.30	3.37	1.78×	2.01
2	GPT-OSS 120B	90.28	3.38	1.53×	1.33
3	QWEN3.5 27B	88.80	2.95	1.36×	1.18
4	GPT-OSS 20B	87.65	3.10	1.34×	2.18
5	QWEN3.5 35B-A3B	86.34	2.74	1.25×	2.33
6	GEMMA-4 26B-A4B-IT	85.91	1.34	1.11×	1.10
7	LLAMA-3.1 8B-INSTRUCT	76.88	1.37	1.06×	3.19
8	OLMO-3 7B-THINK	75.61	5.31	1.28×	0.98
9	OLMO-3 7B-INSTRUCT	66.42	2.83	1.09×	2.53

AI USE STATEMENT

Generative AI was used for benchmark pre-annotation and fictitious entity generation. All generated outputs were reviewed and, where necessary, corrected by human annotators, as detailed in Section 2.2. Generative-AI tools also provided assistance with some software development.

REPRODUCIBILITY STATEMENT

The anonymous dataset and code releases are linked in the abstract. Section 2.1–2.3 describes source selection, human-in-the-loop template annotation, and fictitious-data generation. Appendix E enumerates the released intervention settings; Appendix F records model identifiers, inference settings, hardware, and licenses; Appendix J provides the generation and evaluation prompts; and Appendix C documents evaluation calibration. The pipeline preserves raw outputs, scores, confidence intervals, manifests, and checksums binding each run to its dataset; statistical details are specified in Section 3.

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APPENDIX

A RELATED WORK

A.1 EVALUATING REASONING VIA INTERVENTIONS AND ROBUSTNESS

Several studies investigate the reasoning limits of LLMs by applying controlled interventions to the input text. Prior work perturbs numerical values or alters problem constraints in math and coding datasets to test whether models solve the underlying operation or exploit surface regularities (Stolfo et al., 2023; Hong et al., 2025; Mirzadeh et al., 2025). Similarly, Shrestha et al. (2025) show that logical accuracy degrades when numerical entities are scaled outside standard ranges, and Razeghi et al. (2022) link arithmetic performance to term frequencies in the training data.

These studies show that LLM reasoning can be brittle to input variations. Our objective is narrower: we do not primarily test robustness to arbitrary problem variation, but the dependence of contextual reasoning on parametric familiarity. By swapping factual entities for fictitious ones while preserving the template, the answer expression, and the replacement constraints, we isolate the difference between operating with familiar versus unfamiliar entity anchors.

A.2 FICTITIOUS AND COUNTERFACTUAL BENCHMARKS

To mitigate data contamination and force models to rely on provided context, recent works introduce fictitious documents or counterfactual scenarios (Monteiro et al., 2024; Wu et al., 2024a; Glockner et al., 2025; Modarressi et al., 2024; Wu et al., 2024b). For instance, NeoQA investigates whether LLMs fall back on parametric knowledge when retrieved documents lack sufficient information (Glockner et al., 2025). CofCA provides a step-wise counterfactual multi-hop QA benchmark, making it one of the closest points of comparison for document-level counterfactual reasoning (Wu et al., 2024a).

Previous counterfactual benchmarks nevertheless face a methodological limitation around task parity. Some compare a fictitious dataset against a different factual benchmark, making it difficult to attribute a performance gap strictly to the lack of parametric knowledge rather than to differences in

dataset difficulty (Monteiro et al., 2024). Others regenerate counterfactual passages, which can alter sentence structure, evidence placement, and reasoning path. **MemoReason** is designed to remove that confound: the factual and fictitious settings are paired through the same annotated templates, so the document structure and question logic are held fixed.

The qualitative observation of lower performance on fictitious examples is consistent with Wu et al. (2024b). The paired design changes what can be concluded: because their factual and fictitious conditions use different evidence-question pairs, the gap may also reflect uncontrolled task differences, and failed answers in the fictitious condition have no paired factual answer against which to test direct recall. MemoReason holds the task fixed, allowing the gap to be attributed to entity familiarity within this controlled intervention and enabling the Shortcut Rate analysis.

B TAXONOMY & ANNOTATION

B.1 ENTITY TYPES

The taxonomy defines 14 entity types used to annotate all replaceable spans in the source documents: persons, places, events, military organizations, enterprise organizations, NGOs, government organizations, educational organizations, media organizations, temporals, numbers, awards, legal instruments, and products. Figure 4 summarizes how these entity families connect to the rule layer and the question-answer contract used by MemoReason.

The main reason for introducing this taxonomy is experimental control. MemoReason is not only a benchmark based on fictitious entity replacement: it is designed to replace the familiar anchors that models may have memorized while keeping the document structure, question, answer expression, and reasoning path fixed. For this to work, the replacement operation must know which spans can be swapped independently and which spans must remain coupled. Each entity type therefore has a compact set of attributes that determine what can be replaced and what must stay consistent across mentions. For example, a person entity can expose names, age, gendered forms, nationality, and relationship attributes, while a legal entity can expose both its name and reference code.

We chose a medium-grained taxonomy rather than a single generic entity class or a very fine-grained ontology. A generic class would make replacements syntactically easy but semantically unsafe: replacing an educational organization with a media outlet, or a legal instrument with a product, can preserve surface fluency while breaking the factual role played by the entity in the document. Conversely, an overly fine-grained ontology would make annotation brittle and would create many rare categories that are difficult to replace reliably. The selected types reflect the distinctions that most often affect document validity under replacement: named entities with different social or institutional roles, quantitative and temporal entities governed by constraints, and domain-specific referents such as awards, products, and legal instruments.

B.2 RULES

Rules encode document-specific constraints that must remain true after entity replacement. They include arithmetic relations, compatibility constraints, plural constraints, exact offsets, and value couplings that are required by the text. Generic ordering constraints are handled automatically by the generation pipeline when they preserve only the factual order of numbers, dates, years, or ages. The goal is to avoid brittle fictitious variants where the surface replacement is syntactically valid but the document becomes semantically incoherent.

B.3 ELIMINATING UNIQUE FACTUAL REFERENCES

Some factual documents contain descriptions that uniquely identify the original entity even after direct entity replacement. During annotation, these spans are either kept literal when they are structurally necessary or softened when they would reintroduce a unique factual cue. For example, a formulation such as “the greatest sprinter” can be broadened to avoid forcing the model to reconcile a fictitious name with a world-knowledge fact about a uniquely identifiable person. This step helps ensure that the fictitious setting tests contextual reasoning rather than contradiction handling.

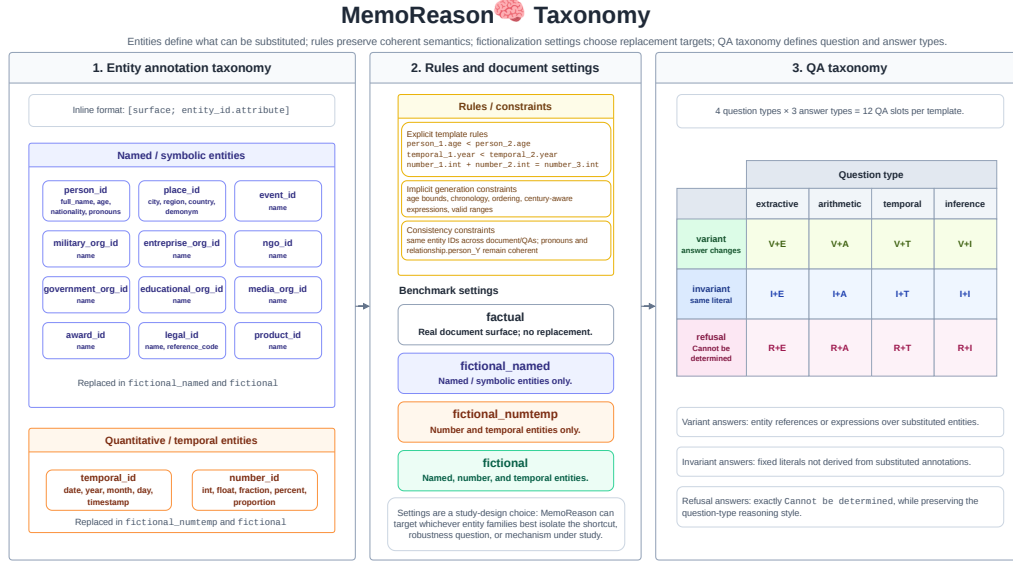


Figure 4: Overview of the **MemoReason** taxonomy and benchmark construction layers. The taxonomy separates semantically distinct entity families, the rule layer preserves document consistency under replacement, and the QA contract crosses question type with answer type.

C JUDGE MATCH CALIBRATION

We model the Judge Match (JM) as a *binary estimator*: given a question, a ground truth answer, and a model prediction as input, it outputs a binary success/failure assessment with respect to the semantic equivalence requirement. Such an estimator is characterized by two recall parameters: its **positive recall** r_+ , i.e., the probability that it correctly identifies a semantically equivalent prediction, and its **negative recall** r_- , i.e., the probability that it correctly identifies a non-equivalent one.

Calibration. To estimate r_+ and r_- , we randomly sampled 200 examples from the training split and had a human annotator label each (question, ground truth, prediction) triple as either a match or a non-match. We then applied JM to the same examples and treated the human annotations as ground truth, yielding $r_+ = 1.0$ and $r_- = 1.0$ (100% agreement). For 200 successes in 200 trials, the two-sided 95% lower bound is 98.2% for each recall, so perfect observed agreement should not be read as zero calibration uncertainty.

Corrected performance estimation. When applying JM to n executions, the raw success rate q (i.e., the fraction of predictions labeled as matches) is a biased estimate of the true performance p . Given the calibrated recalls, we correct for this bias and report p together with a confidence interval using the following formula:

$$p \in \frac{q + r_- - 1}{r_+ + r_- - 1} \pm \Delta_p, \quad \Delta_p = z \frac{\sqrt{\frac{q(1-q)}{n}}}{r_+ + r_- - 1} \quad (1)$$

where z is the quantile of the standard normal distribution corresponding to the desired confidence level (e.g., $z = 1.96$ for a 95% confidence interval). When $r_+ = r_- = 1$, this reduces to $p = q$, confirming that no correction is needed in our case.

D ANNOTATION INTERFACE

We built a custom annotation interface to support the human review loop. The interface is organized around three coupled tasks: document annotation, rule annotation, and question-answer validation. Keeping these tasks in one interface makes it possible for annotators to inspect whether a proposed entity span, rule, or answer expression remains valid under the same template.

Final quality control is layered rather than delegated to a single automatic score. Each template was independently reviewed by two domain-knowledgeable annotators (and by three for a subset), disagreements were adjudicated, and corrections from audit passes were cross-checked by two additional readers together with a sample of unflagged items. One such audit identified 15 subtly problematic questions, approximately 1.4% of the then-current question set; all were corrected and independently reviewed before release.

D.1 DOCUMENT ANNOTATION

Annotators review inline entity annotations over the source document, question, and answer expression. They verify the type of entity, the attribute, and the consistency of repeated references throughout the document.

D.2 RULES ANNOTATION

Annotators then inspect the rule list attached to the template. The goal is not to encode every factual relation, but to encode only the constraints needed for fictitious replacements to preserve the document logic.

D.3 QUESTIONS/ANSWERS ANNOTATION

Each document is paired with 12 question-answer slots, crossing four question types with three answer types. The interface helps annotators verify that each question is answerable from the document when it should be, that refusal questions genuinely lack supporting evidence, and that the answer expression can be evaluated after fictitious replacement.

E DATASET DETAILS

E.1 DATASET THEMES

The factual dataset is organized into nine thematic categories. These themes are not different tasks: all of them are annotated with the same entity taxonomy, replacement rules, question types, and answer types. Their role is to diversify the factual contexts in which we test whether models rely on document-grounded reasoning or parametric shortcuts.

- **Award Winners** contains articles about prominent public figures whose documents include major awards, distinctions, honors, or record-setting achievements. This theme is dense in person, award, organization, event, and temporal entities.
- **Biographies** contains articles about famous personalities, especially scientists, politicians, and institutional leaders. These documents emphasize career trajectories, appointments, offices, affiliations, and chronological progressions across institutions.
- **Places** contains articles about cities, countries, and broader regions. These documents cover geographic, political, demographic, historical, and administrative facts, making them useful for testing place-based relations and numerical comparisons.
- **Companies** contains articles about corporations and organizations. The documents describe founding histories, mergers, acquisitions, subsidiaries, sectors, headquarters, leadership, and market-related facts.
- **Natural Disasters** contains articles about earthquakes, hurricanes, tsunamis, wildfires, and other large-scale disasters. These documents involve event timelines, affected regions, casualties, magnitudes, damages, and institutional responses.
- **Public Attacks** contains WikiEvent-derived news articles about public attacks, threats, and violent incidents.
- **Retail Banking** contains articles about banking regulations, payment systems, deposit guarantees, capital requirements, and resolution mechanisms. This theme introduces legal and institutional language, with reasoning often depending on policy roles, requirements, and timelines.

- **Space Missions** contains articles about spaceflight programs, spacecraft, missions, and space agencies. These documents combine technical mission descriptions with crews, launch dates, mission outcomes, vehicles, and chronological dependencies.
- **Sport Events** contains articles about major competitions and tournaments. These documents include editions, participants, venues, records, audiences, rankings, and event histories, which provide many numerical, temporal, and relational reasoning cases.

E.2 RELEASED DATASETS

We release the ten dataset settings used in the paper experiments. The factual setting is the reference point; every intervention is paired through the same document templates whenever factual and fictitious performance are compared. Public-facing aliases use the `fictitious*` prefix below; frozen experimental manifests retain their original keys for hash compatibility.

Dataset setting	Experiment role	Description
<code>factual</code>	Reference setting	Original factual documents and questions after human template review.
<code>fictitious</code>	Main replacement setting	Full replacement of named, numerical, and temporal entities under the template rules.
<code>fictitious_named</code>	Replacement ablation	Replaces named entities while leaving numerical and temporal values unchanged.
<code>fictitious_numtemp</code>	Replacement ablation	Replaces numerical and temporal entities while leaving named entities unchanged.
<code>fictitious_10pct</code>	Partial replacement	Replaces 10% of eligible entities.
<code>fictitious_20pct</code>	Partial replacement	Replaces 20% of eligible entities.
<code>fictitious_30pct</code>	Partial replacement	Replaces 30% of eligible entities.
<code>fictitious_50pct</code>	Partial replacement	Replaces 50% of eligible entities.
<code>fictitious_80pct</code>	Partial replacement	Replaces 80% of eligible entities.
<code>fictitious_90pct</code>	Partial replacement	Replaces 90% of eligible entities.

Table 6: Released dataset settings used in the experiments. Partial-replacement datasets are generated from the same templates as the factual and fully fictitious settings.

F EXPERIMENTAL SETUP DETAILS

This section reports the inference and scoring settings used for the experiments. Each question is evaluated independently: the model receives the document and a single question, and must return exactly one line of the form `ANSWER: <answer>`. The system prompt is shown in Appendix J.1. We do not request chain-of-thought rationales; when a reasoning-capable model emits hidden or tagged reasoning, the evaluation pipeline stores it separately and parses only the final answer channel or final `ANSWER: line`.

F.1 MODEL INFERENCE SETTINGS

Decoding is deterministic throughout all experiments (i.e. greedy). We also keep reasoning-related controls fixed across runs: GPT-OSS API requests use `reasoning_effort=low` and do not request reasoning traces, while local GPT-OSS runs use the corresponding low-reasoning setting in the chat template.

F.2 HARDWARE

Experiments were performed using NVIDIA H100 and A100 GPUs with 80 GB of VRAM. Generating the local open-weight model outputs, including ablations, required approximately 200–250 GPU hours. This estimate excludes provider-side compute for API-hosted models.

F.3 MODEL LICENSES

The open-weight models evaluated in this paper are used under the licenses stated by their providers. GPT-OSS-20B and GPT-OSS-120B are released under the Apache 2.0 license (OpenAI et al., 2025). Qwen3.5-27B and Qwen3.5-35B-A3B are released under the Apache

2.0 license (Qwen, 2026). The OLMo-family models, including OLMo-3-7B-Instruct, OLMo-3-7B-Think, and OLMo-2-32B-Instruct, are released under the Apache 2.0 license (Olmo et al., 2026). Gemma-4-26B-A4B-it is released under the Gemma license terms available at https://ai.google.dev/gemma/apache_2. Llama-3.1-8B-Instruct is used under the Llama 3.1 Community License available from Meta at https://www.llama.com/llama3_1/license/. For API-only systems, no model weights are redistributed by this work: Claude Sonnet 4.6 and Opus 4.6 are governed by Anthropic’s service terms (Anthropic, 2026b;a).

G FULL RESULTS

G.1 ROBUSTNESS AND MECHANISM CHECKS

The following analyses consolidate the endpoint result without changing its estimand. The cue ablation separates named-entity identity from numerical/temporal values; directional flips test the asymmetry item by item; and the reverse intervention tests factual-answer copying while familiar names remain in the context. Each reported confidence interval resamples complete source documents.

G.1.1 NAMED-ENTITY AND VALUE CUES

Table 7: Accuracy over the 400 invariant-answer questions spanning arithmetic, temporal, inference, and extractive questions. Each non-factual cell reports accuracy and, below it, the setting-minus-factual change with its 95% CI. * marks an interval excluding zero.

Model	Factual	Values only	Names only	Full fictional
OLMO-3 7B-INSTRUCT	58.25	57.65 $\Delta = -0.60 [-3.25, +2.08]$	54.47 $\Delta = -3.78 [-6.73, -0.83]^*$	53.65 $\Delta = -4.60 [-7.72, -1.52]^*$
OLMO-3 7B-THINK	74.50	73.75 $\Delta = -0.75 [-4.08, +2.65]$	73.17 $\Delta = -1.33 [-4.92, +2.37]$	71.70 $\Delta = -2.80 [-6.30, +0.72]$
QWEN3.5 35B-A3B	86.50	82.05 $\Delta = -4.45 [-6.57, -2.43]^*$	84.77 $\Delta = -1.73 [-3.98, +0.50]$	82.55 $\Delta = -3.95 [-6.73, -1.32]^*$

Across invariant-answer arithmetic, temporal, inference, and extractive questions, the three evaluated models do not share a single necessary cue class. OLMo-3-7B-Instruct is significantly affected when names are replaced but not by values alone; Qwen3.5-35B-A3B shows the opposite pattern. Neither isolated cue intervention is significant for OLMo-3-7B-Think. This model dependence argues against reducing the familiarity effect to one universal surface cue.

G.1.2 DIRECTIONAL ITEM-LEVEL FLIPS

Table 8: Directional flips across all question and answer classes; mirror is $P(\text{factual incorrect} \mid \text{fictional correct})$.

Model	Forward flip (%)	Mirror flip (%)	Forward — mirror (pp)
OLMO-3 7B-THINK	16.29 [14.60, 18.02]	11.24 [9.67, 12.90]	5.05 [2.82, 7.25]
OLMO-3 7B-INSTRUCT	15.37 [13.74, 17.03]	10.97 [9.22, 12.77]	4.40 [2.15, 6.63]
GPT-OSS 20B	7.17 [5.89, 8.51]	4.75 [3.57, 6.01]	2.42 [0.61, 4.25]
GEMMA-4 26B-A4B-IT	5.53 [4.33, 6.80]	3.41 [2.51, 4.38]	2.12 [0.57, 3.68]
GPT-OSS 120B	5.50 [4.29, 6.78]	3.08 [2.24, 3.96]	2.42 [0.92, 3.97]
QWEN3.5 27B	4.89 [3.73, 6.15]	2.53 [1.79, 3.32]	2.36 [0.95, 3.80]
QWEN3.5 35B-A3B	5.79 [4.53, 7.17]	2.71 [1.94, 3.56]	3.08 [1.57, 4.66]
CLAUDE SONNET 4.6	4.66 [3.62, 5.77]	3.16 [2.25, 4.17]	1.50 [0.06, 2.94]
LLAMA-3.1 8B-INSTRUCT	8.03 [6.52, 9.65]	7.09 [5.65, 8.58]	0.94 [-1.27, 3.17]

Our primary directional analysis follows the full benchmark scope and includes every question and answer class. Forward flips—factual successes that become failures in the fictitious setting—significantly exceed the mirror rate for 8 of 9 models (Table 8); the remaining difference is inconclusive.

G.1.3 REVERSE INTERVENTION WITH FACTUAL NAMES RETAINED

Strict reproduction of the original factual answer remains rare when numerical and temporal values are changed but familiar names are retained: aggregate reasoning Shortcut Rates are 1.95–3.22% across the 4 evaluated models. The largest atomic cell reaches 5.00%, a localized exception rather than a general strategy.

Table 9: Shortcut Rate under value-only replacement (%; failed outputs in parentheses).

Model	Arith.	Temp.	Infer.	Reason.
OLMO-3 7B-INSTRUCT	3.80 (612)	3.44 (547)	1.86 (375)	3.03 (1534)
OLMO-3 7B-THINK	2.70 (358)	2.92 (298)	0.25 (198)	1.95 (854)
QWEN3.5 35B-A3B	3.40 (328)	1.25 (303)	5.00 (194)	3.22 (825)
CLAUDE SONNET 4.6	3.83 (139)	2.50 (70)	2.20 (98)	2.84 (307)

G.1.4 DECODING TEMPERATURE

Table 10: **Decoding-temperature robustness.** Accuracy (%) and factual–fictitious gap (percentage points). Brackets are 95% CIs. Nonzero temperatures average three generation seeds; $T = 0$ uses the deterministic run. Bold gaps have an interval excluding zero.

Model	T	Factual acc.	Fictional acc.	Gap (pp)
OLMo-3 7B-Instruct	0.0	69.67 [67.42, 71.92]	66.48 [64.55, 68.38]	3.19 [1.35, 5.02]
		69.42 [67.03, 71.78]	66.02 [64.15, 67.87]	3.40 [1.62, 5.16]
	0.5	68.81 [66.42, 71.19]	65.33 [63.47, 67.16]	3.48 [1.69, 5.27]
Qwen3.5 27B	0.0	91.67 [90.17, 93.17]	88.78 [87.13, 90.38]	2.88 [1.43, 4.35]
		91.61 [90.14, 93.06]	88.69 [87.10, 90.24]	2.92 [1.52, 4.35]
	0.5	91.00 [89.44, 92.53]	88.06 [86.49, 89.61]	2.94 [1.54, 4.37]

Across the 2 evaluated models, the factual–fictitious gap remains positive and its paired 95% interval excludes zero in all 6 model–temperature cells. Increasing temperature does not produce a detectable change in the gap: relative to $T = 0$, the changes range from 0.03 to 0.29 points, and all 4 nonzero-temperature paired contrast intervals contain zero. The effect therefore persists under stochastic decoding in this targeted check, although this focused sweep does not establish broad temperature invariance.

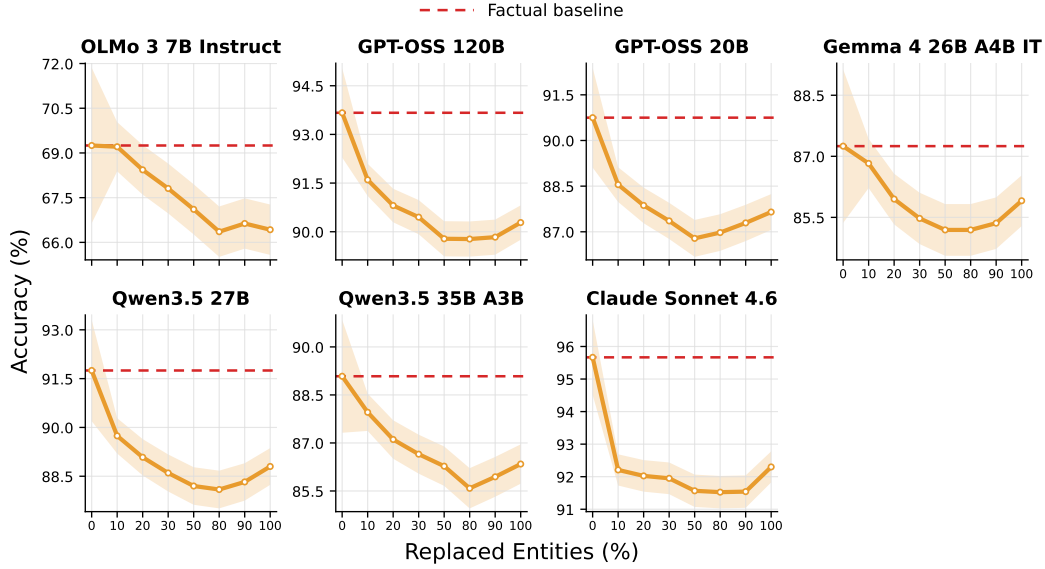


Figure 6: Model accuracy across exact replacement proportions. All seven panels show the complete sweep. Dashed red lines show factual performance; shaded bands are 95% CIs.

G.1.5 TRAINING-DATA EXPOSURE

We test whether the endpoint drop can be explained by how often a document’s replaced named entities occur in the OLMo training mix. For each source document, we average historical Infini-gram counts over its replaced named entities and relate this quantity to the paired factual-to-fictitious drop. The estimated association is weakly negative: Pearson $r = -0.103$ (95% CI $[-0.252, 0.028]$), Spearman $\rho = -0.121$ ($[-0.298, 0.064]$), and the fitted slope is -2.34 points per frequency decade ($[-6.42, 0.66]$). All intervals resample source documents and include zero. Thus, exposure frequency alone does not reliably account for the familiarity effect.

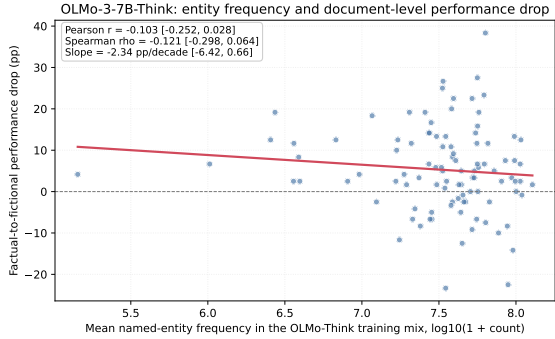


Figure 5: OLMo-3-7B-Think document-level drop versus mean named-entity frequency in its training mix. The line is an OLS fit; brackets in the inset are 95% CIs.

G.2 PARTIAL REPLACEMENTS

Figure 6 shows the 7 complete partial-replacement sweeps. Six curves recover clearly from a 50%–80% minimum toward the fully fictitious endpoint, by 0.51–0.87 points; OLMo-3-7B-Instruct changes by only 0.07 points and is better described as plateauing. Thus, the mixed-context pattern is common but not universal in magnitude, and the overlapping confidence intervals counsel against treating it as causal evidence by itself.

The same figure includes the complete partial-replacement sweep for Claude Sonnet 4.6. Accuracy falls from 95.67% in the factual setting to 91.53% at 80% replacement, then recovers by 0.78 points at the fully fictitious endpoint.

H QUALITATIVE ANALYSIS

H.1 QUALITATIVE THINKING-CHECKPOINT CONTRAST OVER FICTITIOUS CONTEXTS

Qualitative analysis: chain-of-thought helps fictitious reasoning

Question. How many years passed between the liberation of Velanthor and the formation of the Sovereign Pact Stability Council ?

Relevant document evidence. [...] Following its liberation in 1938, the short-lived Tessarine Commonwealth was established and later dissolved over the course of the Telvoran Conflict. [...] The current Sovereign Pact Stability Council, a semi-presidential system, was formed in 1949 by Tarvien Molquesp.

Gold answer. $1949 - 1938 = 11$

OLMo-3-7B-Instruct

Model response: ANSWER: 9

Outcome: Incorrect.

OLMo-3-7B-Think

Model chain-of-thought and answer:

Okay, let's see. The question is asking how many years passed between the liberation of Velanthor and the formation of the Sovereign Pact Stability Council.

First, I need to find the dates mentioned in the document for both events. Let's start with the liberation. The document says, "Following its liberation in 1938 ..." So the liberation happened in 1938.

Now, the Sovereign Pact Stability Council was formed in 1949. Wait, the exact phrase is "formed in 1949 by Tarvien Molquesp." Hmm, the year here is written as 1949. That might be a typo or maybe it's supposed to be 1949? Usually, years are four digits, so 1949 makes sense as 1949. So the council was formed in 1949, which I'll take as 1949.

So the time between 1938 and 1949. Let me calculate the difference. From 1938 to 1949 is 11 years. Because $1949 - 1938 = 11$.

Wait, but I should check if there's any other dates involved. The liberation is clearly 1938. The formation of the council is 1949. The question is straightforward subtraction. The answer should be 11 years.

I need to make sure there's no other events in between that might affect the count. The document mentions the collapse in 1931, then the Commonwealth in 1938? Wait no, let me re-read that part.

The liberation was in 1938. Then the next part says "the short-lived Tessarine Commonwealth was established and later dissolved over the course of the Telvoran Conflict." The Telvoran Conflict's timeline isn't given, but the next event mentioned is the formation of the Sovereign Pact in 1949. So between 1938 and 1949, that's 11 years.

Yes, so 1949 minus 1938 is 11. The answer is 11. The document doesn't mention any other years in between, so I think that's it. Just subtract the two years given. So the answer should be 11.

Final answer: ANSWER: 11

Outcome: Correct.

I LIMITATIONS AND RESPONSIBLE RELEASE

Limitations. MemoReason is designed for controlled document-grounded reasoning experiments, not as a complete model-understanding probe. The benchmark is English-only, and the reported

conclusions are limited to the selected model set and prompting protocol. Its source-document scale and nine-theme coverage support controlled paired inference but not exhaustive theme- or entity-level generalization; reported uncertainty accounts for this dependence. Main evaluations use greedy decoding, while Appendix G.1.4 provides a targeted two-model temperature check.

Responsible release. The release contains benchmark examples and evaluation code, not a trained model. We anonymize publication identifiers, remove annotator names from released data, include a canary field in dataset rows, and publish the dataset as explicit splits so users can distinguish factual, fictitious, partial-replacement, and temporal-perturbation settings. The main intended use is diagnostic evaluation of contextual faithfulness; because the dataset contains fictitious passages, users should avoid presenting individual fictitious documents as real-world claims.

J PROMPTS

J.1 EVALUATION SYSTEM PROMPT

Evaluation System Prompt

You answer questions using only the provided document.
 Rules:
 - Use only the document. Do not use outside knowledge.
 - Output exactly one line in the format: ANSWER: <answer>
 - Do not explain your reasoning.
 - Do not output a calculation or equation.
 - Do not restate the question.
 - Do not add any extra text before or after the answer line.
 - If the document does not determine the answer, output exactly:
 ANSWER: Cannot be determined
 - Otherwise, return only the shortest final answer.

J.2 TAXONOMY SPECIFICATION PROMPT

Taxonomy Specifications

List of entity types with their associated attributes:

- person_ID
- place_ID
- event_ID
- military_org_ID
- enterprise_org_ID
- ngo_ID
- government_org_ID
- educational_org_ID
- media_org_ID
- temporal_ID
- number_ID
- award_ID
- legal_ID
- product_ID

Where ID is a unique identifier that is given to each entity type instance that is mentioned in the document. This identifier should be used consistently across all the references of the same entity in the document.

List of attributes for each entity type:**person_ID**

(Individual human being mentioned in the document)

- **full_name**: Complete name of the person (first, middle, and/or last). Example: [John Smith; person_1.full_name], [Marie Curie; person_1.full_name]
- **first_name**: Person's given name. Example: [John; person_1.first_name], [Marie; person_1.first_name]
- **last_name**: Person's family name. Example: [Smith; person_1.last_name], [Curie; person_1.last_name]
- **age**: Person's age in years. Example: [25; person_1.age], [45; person_1.age]
- **gender**: Person's gender. Example: [male; person_1.gender], [female; person_1.gender]
- **nationality**: Person's nationality or citizenship. Example: [American; person_1.nationality], [French; person_1.nationality]
- **ethnicity**: Person's ethnic or cultural background. Example: [Hispanic; person_1.ethnicity], [Asian; person_1.ethnicity]
- **subj_pronoun**: Subject pronoun (he/she/they). Example: [he; person_1.subj_pronoun], [she; person_1.subj_pronoun]
- **obj_pronoun**: Object pronoun (him/her/them). Example: [him; person_1.obj_pronoun], [her; person_1.obj_pronoun]
- **poss_det_pronoun**: Possessive determiner (his/her/their). Example: [his; person_1.poss_det_pronoun], [her; person_1.poss_det_pronoun]
- **poss_pro_pronoun**: Possessive pronoun (his/hers/theirs). Example: [his; person_1.poss_pro_pronoun], [hers; person_1.poss_pro_pronoun]
- **refl_pronoun**: Reflexive pronoun (himself/herself/themselves). Example: [himself; person_1.refl_pronoun], [herself; person_1.refl_pronoun]
- **honorific**: Title or honorific. Example: [Mr.; person_1.honorific], [Ms.; person_1.honorific]
- **relationship**: Special attribute for relationships between people (use relationship.person_Y). Example: [mother; person_1.relationship], [brother; person_1.relationship]
- **middle_name**: Person's middle name. Example: [Marie; person_1.middle_name], [Fitzgerald; person_1.middle_name]

place_ID

(Geographic location or landmark)

- **city**: Name of a city or town. Example: [New York; place_1.city], [Paris; place_1.city]
- **region**: Geographic region or area. Example: [New England; place_1.region], [Midwest; place_1.region]
- **state**: State or province. Example: [California; place_1.state], [Ontario; place_1.state]
- **country**: Nation or country. Example: [United States; place_1.country], [France; place_1.country]
- **street**: Street name or address. Example: [Main Street; place_1.street], [5th Avenue; place_1.street]
- **natural_site**: Natural landmark or feature. Example: [Mount Everest; place_1.natural_site], [Amazon River; place_1.natural_site]
- **continent**: An entity referring to a continent. Example: [Europe; place_1.continent], [Africa; place_1.continent]
- **demonym**: The demonym of a place entity. Example: [Syrian; place_1.demonym], [African; place_1.demonym]

event_ID

(Named event or occurrence)

- **name**: Name of the event. Example: [World War II; event_1.name], [Olympic Games; event_1.name]
- **type**: Type or category of event. Example: [war; event_1.type], [conference; event_1.type]

military_org_ID

(Military organization)

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- name: Name of the military organization. Example: [IRA; military_org_1.name], [Royal Guard Command; military_org_1.name]

enterprise_org_ID

(Private/company organization (enterprise))

- name: Name of the enterprise organization. Example: [Google; enterprise_org_1.name], [BNP Paribas; enterprise_org_1.name]

ngo_ID

(NGO / non-governmental organization)

- name: Name of the NGO. Example: [Amnesty International; ngo_1.name], [Médecins Sans Frontières; ngo_1.name]

government_org_ID

(Government institution or agency)

- name: Name of the government organization. Example: [U.S. State Department; government_org_1.name], [European Commission; government_org_1.name]

educational_org_ID

(Educational institution)

- name: Name of the educational organization. Example: [Harvard University; educational_org_1.name], [Sorbonne Université; educational_org_1.name]

media_org_ID

(Media/journalism organization)

- name: Name of the media organization. Example: [BBC; media_org_1.name], [Reuters; media_org_1.name]

temporal_ID

(Time-related information (dates, times, etc.))

- day: Day of the week. Example: [Monday; temporal_1.day], [Friday; temporal_1.day]
- date: Full date. Example: [January 1, 2024; temporal_1.date], [2024-01-01; temporal_1.date]
- year: Year. Example: [2024; temporal_1.year], [1999; temporal_1.year]
- month: Month name or number. Example: [January; temporal_1.month], [12; temporal_1.month]
- timestamp: Specific time or timestamp. Example: [3:30 PM; temporal_1.timestamp], [15:30; temporal_1.timestamp]
- day_of_month: Day number within a month. Example: [15; temporal_1.day_of_month], [1st; temporal_1.day_of_month]

number_ID

(Numerical values and quantities)

- int: Integer number. Example: [42; number_1.int], [100; number_1.int]
- str: Number expressed as text. Example: [forty-two; number_1.str], [one hundred; number_1.str]
- float: Decimal number. Example: [3.14; number_1.float], [99.9; number_1.float]
- fraction: Fractional value. Example: [1/2; number_1.fraction], [three quarters; number_1.fraction]
- percent: A number representing a percentage (must be between 0 and 100). Example: [20; number_1.percent], [10; number_1.percent]
- proportion: A number that represents a proportion (must be between 0.0 and 1.0). Example: [0.1; number_1.proportion], [0.00012; number_1.proportion]

award_ID

(An entity that represents any kind of awards)

- name

legal_ID

(Legal or regulatory instrument (law, directive, regulation, treaty, policy framework))

- name: Legal or regulatory instrument (law, directive, regulation, treaty, policy framework). Example: [Markets in Financial Instruments Directive 2014; legal_1.name], [Capital Requirements Regulation; legal_1.name]
- reference_code: Official legal citation or identifier. Example: [2014/65/EU; legal_1.reference_code], [(EU) No 575/2013; legal_1.reference_code]

product_ID

(An entity describing a product)

- name: The name of the product as mentioned in the text. Example: [iPhone; product_1.name], [Eliquis (apixaban); product_1.name]

J.3 ENTITY ANNOTATION PROMPT**Entity Annotation Prompt**

```
# Entity Annotator
You are an annotation agent. Annotate the document, questions, and
answers below using inline annotations only.
Output format:
Render the annotated document, questions, and answers in a YAML
form with the following structure:
document_id: <doc_id>
annotated_document: |
  <full document with inline annotations [span; entity.attr]>
annotated_questions:
  - question_id: <id>
    question: <question with inline annotations>
    question_type: <extractive|arithmetic|temporal|inference>
    answer: <answer_expression>
decision_log:
  - span: "<text span as it appears in document>"
    action: ANNOTATE | KEEP_LITERAL
    reason_code: SAFE_TO_TEMPLATE | OUT_OF_TAXONOMY |
STRUCTURAL_CRITICAL_SPAN | CORE_SEMANTIC_CRITICAL_SPAN |
COUPLED_DEPENDENCY | AMBIGUOUS
    short_reason: "<one short sentence>"
Here are a few rules to follow when annotating the document,
questions, and answers:
1. Do NOT modify the document text except by inserting inline
annotations.
2. Do NOT modify question/answer text except by inserting inline
annotations.
3. Inline annotation format MUST be exactly: [span_text;
entity_id.attribute]
4. After removing annotations (keeping only span_text), the text
must be identical to the original document.
5. Use only the taxonomy below. If an annotation would violate it,
do NOT annotate it.
{{TAXONOMY_SPECIFICATIONS_PROMPTJ.2}}
```

Figure 7: The prompt used to draft AI Agent pre-annotations to help human annotators.

J.4 RULE GENERATION PROMPT**Rule Generation Prompt**

```
Your are a rigorous rule generation agent and your task is to
generate a list of rules based on an annotated document.
Purpose of these rules:
- The annotated template will later be converted into factual and
fictional documents.
- During fictionalization, the code replaces annotated entities with
sampled fictional ones (i.e. non existing entities) and
```

regenerates numbers, temporals, and gender-related attributes (e.g. pronouns, honorific, etc.).

- The problem is that these new fictional/non-existing entities may not be suitable for the context of the document, we therefore need to define sampling rules/constraints.
- Your task is to produce logical constraints (rules) that will be used to sample fictional entities to make sure that semantic/linguistic coherence is still maintained accross the document after the replacement.
- The rule list tells the generator which document-specific semantic constraints must still hold after replacement.
- Basically those rules serve as safe guards to avoid replacing with fictional entities that break the semantic and coherence of the document.
- Without these rules, the generator could produce a fictional document that is no longer coherent.

How to think about the task:

- You are identifying the minimal set of explicit constraints that must be satisfied by the entity replacement.
- A good rule protects a fact that is stated or directly implied by the document and that could otherwise break after replacement.
- Focus specifically on the annotated entities (i.e. text spans between brackets with an entity type, ID, and attribute [text span; entity_type_ID.attribute] - a detailed description of the entity taxonomy is provided below.)

General notes:

- Use only entities and attributes that are actually annotated in the document.
- The code performs automatic generation for numbers, temporals, age, and gendered forms (pronouns, gender, honorific, etc.) and sampling from a pre-defined pool for the rest of attributes such as names.
- Do not write rules whose only purpose is to preserve order for numbers, ages, dates, or years. This is already automatically handled by the code.
- Never write pure chronology/order rules between temporal entities (for example 'temporal_2.year < temporal_3.year').
- Never write pure order rules between number entities (for example 'number_1.int < number_2.int') or age entities (for example 'person_1.age < person_2.age') when the goal is only to preserve factual ordering.
- Important: if the document implies an exact gap/offset, you should encode that relation explicitly (for example consecutive years: 'temporal_3.year - temporal_2.year == 1').
- Century mentions are annotated as 'number' entities, not 'temporal' entities. When a century mention is semantically tied to an exact year/date and changing them independently could create an obvious contradiction, add an explicit century rule.
- Allowed century helper functions:
 - 'century_of(temporal_X.year)' or 'century_of(temporal_X.date)' when an exact year/date must stay inside a stated century
 - 'century_start(number_X.int)' and 'century_end(number_X.int)' when a stated century must stay before/after an exact year/date
- Typical century rules:
 - 'century_of(temporal_6.year) == number_8.int'
 - 'century_end(number_9.int) < temporal_7.year'
- Crucial: Never use the factual values of entities but only use the actual entity reference, for example in the sentence *They reported [9; number_3.int] deaths at the Oxford Avenue* never use the factual value 9 explicitly but rather use its associated reference 'number_3.int'. This applies to all the other entities/attributes such as temporals and 'person.age'. This also applies to sums, don't force them to sum to a certain factual fixed

value, but rather they must sum to the entity reference that is used to annotate the factual value. For example in the sentence *There were [9; number_3.int] deaths and [3; number_4.int] injured resulting in [12; number_5.int] casualties.* never use the factual value 12 to represent the result of summing 'number_3.int' and 'number_4.int' but rather use its reference 'number_5.int' ('number_3.int + number_4.int == number_5.int').

- Constants are allowed when they encode a structural relation (for example '== 1' for consecutive years).
- Don't fix entities to a have specific value that is mentioned in the text (e.g. 'number_10.int == 10'), remember that we actually want to replace entities with different values so fixing them does make sense.
- add a small concise comment next to each rule to quickly explain why, using inline '#' format (e.g. 'number_5.int > 1 # plural mention')

What a good rule captures:

- an arithmetic relation that must stay true
- an explicit comparison or bound
- a document-specific dependency that would become incoherent if broken
- a semantic constraint whose violation would make the fictionalized document inconsistent
- an explicit interval/offset equation between temporals, numbers, or ages when the text implies a specific gap
- a sum of numbers, temporals, or ages that must sum to a certain value
- a century/year compatibility rule when a stated century must remain compatible with an exact year or date
- if you notice that two entities are the same and should have the same values but are annotated with different entity IDs then indicate this with a rule (e.g. 'number_2.int == number_3.int')
- a rule that indicates that a number should be plural because there is a plural mention in the text (e.g.)

What should not become a rule:

- generic number ordering
- generic temporal ordering
- string-content constraints over name fields
- repeated-value matching just because two annotations happen to share the same factual value (you have to make sure that they are the same entities because sharing the same value doesn't mean that it's the same entities)
- date decomposition rules that only restate that a full date includes its year or month
- biography chronology chains that simply mirror the order in which events happened
- anything based on outside knowledge
- vague or speculative constraints
- trivial restatements of the annotations
- facts that the generation code already enforces automatically

Bad rule patterns to avoid:

- 'temporal_7.year < temporal_8.year'
- bad because it only encodes chronology.
- 'number_1.int < number_2.int'
- bad because it only preserves factual ordering.
- 'person_1.age < person_2.age'
- bad because it only preserves factual ordering.

When to return no rule: if the only candidate rules are chronology/order rules, string-content rules over names, repeated-value rules, or date/number decomposition rules, return 'rules: []'.

In [2022; temporal_4.year], [Nora; person_1.name], who was [6; person_1.age] years old, joined the [Oak Street kids' science;

```

entreprise_org_1.name] club. In [2023; temporal_5.year], a year
after [she; person_1.subj_pronoun] joined the club, [her;
person_1.obj_pronoun] older [brother;
person_2.relationship.person_1] [Sam; person_2.name] joined too.
[Sam; person_2.name] was [8; person_2.age] years old, so [he;
person_1.subj_pronoun] was [2; number_7.int] years older than [Nora;
person_1.name]. During the spring tournament, the club won [3;
number_2.int] robotics rounds and [2; number_3.int] quiz rounds,
for a total of [5; number_4.int] wins. At the summer fair, [Sam;
person_2.name] arrived with [4; number_5.int] friends; together
with the [3; number_2.int] club children already waiting, their
group included [7; number_6.int] children. The club archives say a
reform charter was written in the [19th; number_8.int] century and
officially adopted in [1875; temporal_6.year]. A related dispute
started in the [18th; number_9.int] century and was settled in
[1802; temporal_7.year].

```

Examples:

```

- 'number_2.int + number_3.int = number_4.int'
- 'temporal_5.year - temporal_4.year == 1 # consecutive calendar
  years'
- 'person_2.age - person_1.age == number_7.int # age gap stated in
  the document'
- 'number_5.int > 1 # plural mention'
- '4 < person_2.age < 12 # age constraints for specific profiles,
  here for example the text mentions that a person is a child and
  kids'
- 'number_6.int = 2 + number_5.int'
- 'century_of(temporal_6.year) == number_8.int # the adoption year
  must stay inside the stated century'
- 'century_end(number_9.int) < temporal_7.year # the settlement year
  must come after the stated starting century'

```

Inputs you will receive:

```

- 'document_id'
- 'annotated_document'
OUTPUT FORMAT (STRICT YAML ONLY):
``yaml
document_id: <doc_id>
rules:
- <rule_expression> # <short_comment>

```

J.5 FICTITIOUS ENTITY POOL GENERATION PROMPT

The fictitious entity pool prompt instructs the generation agent to produce document-specific pools of non-existing named entities, while leaving numbers, dates, years, pronouns, and other automatically generated fields to the deterministic generator. Prompts are reproduced verbatim; their original terminology is retained to preserve the exact experimental record.

Fictitious Pool Generation Prompt

You generate document-specific pools of fictional named-entity variants for benchmark construction.

Purpose:

- The benchmark replaces factual entities with fictional ones and then generates fictional document versions from those pools.
- Each required entity reference must receive its own list of fictional variants.
- The same document variant index is shared across all references generated in one response.
- If rules connect several references in this batch, variant '1' for all of those references must jointly satisfy the rules, variant '2' must jointly satisfy the rules, and so on up to variant '15'.
- Your output is not the final fictional document. It is the reference-specific candidate pool that later code will sample from.

What must stay outside the pool:

- Do NOT generate numbers.
- Do NOT generate dates, years, weekdays, months, timestamps, or any other temporals.
- Do NOT generate person gender attributes, pronouns, honorifics, or relationship fields.
- Those values are generated later by Python code with fixed seeds and rule checking.

Why the rules matter here:

- The rules describe constraints that the later document generator must preserve.
- Your pool should make those constraints easy to satisfy.
- Example: if a rule requires `'person_1.nationality == place_2.demonym'`, generate person nationalities and place demonyms that can be matched cleanly.
- Example: if a legal item needs a `'name'` and a `'reference_code'`, generate values that plausibly belong to the same fictional legal instrument.
- Ignore rules that only concern numbers, temporals, or other automatically generated fields.
- When you invent a place and a corresponding demonym or nationality adjective, they must belong to the same invented place.
- Do not pair a country, city, region, or state with a demonym that clearly belongs to some other invented place.
- If a place variant contains both `'demonym'` and `'nationality'`, keep them aligned unless the document explicitly requires different surface forms.

Entity taxonomy reference for this prompt:

- The list below is injected from the project taxonomy so the entity meanings and attribute meanings match the rest of the codebase exactly.
- ```
{{ENTITY_TAXONOMY_REFERENCE}}
```

Pool construction requirements:

- Every invented value must be fictional.
- In this benchmark, "fictional" means "non-existing": the value must not refer to a real entity that already exists.
- Treat any candidate that appears on Wikipedia or in a web search as invalid for this task.
- Before finalizing the pool, explicitly check whether the names you generated correspond to real entities. If your environment gives you web search or browsing tools, use them. If it does not, regenerate any candidate that seems plausibly real or widely used.
- Do NOT use ordinary attested first names, surnames, city names, country names, demonyms, award names, legal names, or organization names.
- Common human names are invalid even if you combine them with other fictional fields.
- If a candidate looks like a standard French, English, Spanish, Portuguese, German, Italian, Arabic, Slavic, or otherwise attested real-world name, reject it and invent a new one.
- Bad examples of invalid outputs: `'Eliane'`, `'Lucien'`, `'Margaux'`, `'Renaud'`, `'Marcelo'`, `'Gaston'`, `'Adrienne'`, `'Henrik'`, `'Mallaby'`, `'Stembridge'`, `'Redwick'`, `'Marbleton'`.
- Good outputs should feel pronounceable but unattested: they should read like plausible names while still looking clearly invented.
- Every invented name must be pronounceable for an English speaker.
- Follow ordinary English phonotactics. Avoid impossible consonant clusters, unreadable punctuation, excessive doubled letters, and fantasy-style spellings.
- Use ASCII only.
- Avoid accents, diacritics, emoji, and decorative punctuation.
- Avoid names that obviously match real well-known entities.
- Avoid names that are only tiny edits of famous real entities.
- Avoid slight edits of ordinary real first names or surnames.
- Avoid ordinary English surname or town-style endings such as `'-ton'`, `'-bridge'`, `'-wick'`, `'-bury'`, `'-ford'`, or `'-by'` when the result looks like an attested real place or family name.
- Avoid slight edits of real demonyms, historical labels, dynasties, eras, empires, or treaty names.
- Do not use standalone Roman numerals or ordinal dynastic labels as entity names.
- Do not invent names by taking a real root and adding a thin suffix like `'-an'`, `'-ian'`, `'-ish'`, `'-ic'`, or `'-a'`.
- Do not use generic suffix-marker tokens such as `'Alt'`, `'Astra'`, `'Nova'`, `'Prime'`, or `'Sigma'` anywhere in generated values.
- Avoid joke names, placeholders, and nonsense strings.
- Keep the pool diverse. Do not output many near-duplicates that only differ by one letter or one generic suffix.
- If a place entry contains several attributes, they must be internally coherent inside the same object.
- If a place entry contains both a place name and a demonym, the demonym must clearly match that exact fictional place.

- If an organization entry is tagged with a subtype, the name must sound plausible for that subtype.

Semantic consistency guardrails (critical):

- Do not generate names that semantically contradict explicit cues in the document text.
- Preserve ideological/polarity cues when those cues are explicitly stated.
- Avoid lexical markers that imply the opposite of the described role.
- If the text gives a clear stance, mission, alignment, or institutional function, generated names must remain compatible with that context.
- Use these examples as strict constraints:
  - If a political party is described as left-wing/progressive, do not generate a name that strongly signals right-wing/ultra-conservative/monarchist alignment.
  - If a party is described as conservative/right-wing, do not generate a name that strongly signals socialist/leftist alignment.
  - If an organization is described as humanitarian, relief-focused, or pacifist, do not generate a militaristic or combat-framed name.
  - If an organization is described as environmental/climate-focused, do not generate a name that suggests fossil-fuel expansion or anti-environment positioning.
  - If an entity is a court, ministry, or regulator, do not generate a name that sounds like a private company brand.
  - If an entity is a company/commercial operator, do not generate a name that sounds like a government ministry or tribunal.
  - If an entity is a university/school, do not generate a name that sounds like a bank, military unit, or political party.
  - If a media outlet is described as local/regional, do not generate a name implying global or official state-agency status unless the text supports it.
  - If the document is about countries, empires, wars, revolutions, or historical periods, do not output names that look like real-world historical labels or near-variants such as 'Frankish', 'Gallican', 'Europan', 'Bourbon Restoration', 'Golden Era', or bare numerals like 'III'.

Reference-level requirements:

- allocate `{{TARGET_CANDIDATES_PER_ENTITY}}` fictional candidates for each unique entity reference requested below.
- Generate exactly `{{TARGET_CANDIDATES_PER_ENTITY}}` fictional variants for every required entity reference shown below.
- Treat the requested reference ids as opaque keys for this call. Do not rename them, renumber them, substitute different ids from the document, or output a nearby reference id.
- If this call requests 'place\_7' and 'place\_8', then output exactly 'place\_7' and 'place\_8'. Outputting 'place\_2', 'place\_4', or any other reference id is invalid.
- Do not merge entity references together, even when they share the same type.
- Keep variants globally distinct across entity references of the same bucket. Do not reuse the same fictional person/place/event/etc. under two different reference ids.
- Only include the attributes required for that specific reference. Do not add unrelated optional attributes.
- For organization entities, write values into the taxonomy bucket that matches the annotation type.
- Within one response, list order matters: 'variants[0]' across linked references describes one coherent fictional document version, 'variants[1]' describes another, and so on.
- When linked references involve nationality or demonym fields, keep the same variant index aligned so the person/place pair still matches at that index.

Inputs:

- 'document\_id': `{{DOCUMENT_ID}}`
- 'document\_theme': `{{DOCUMENT_THEME}}`
- 'annotated\_document\_excerpt\_for\_this\_call': `{{ANNOTATED_DOCUMENT}}`
- 'requested\_reference\_mentions': `{{REQUESTED_REFERENCE_MENTIONS}}`
- 'rules\_relevant\_to\_pool\_generation': `{{POOL_RELEVANT_RULES}}`
  - Treat this as the actionable rule subset for pool construction.
- 'required\_entities\_summary': `{{REQUIRED_ENTITIES_SUMMARY}}`
- 'reference\_ids\_for\_this\_call': `{{REFERENCE_IDS_FOR_THIS_CALL}}`
- 'reference\_target\_counts': `{{REFERENCE_TARGET_COUNTS}}`

```

Output format (STRICT YAML ONLY):
```yaml
persons:
  person_1:
    required_attributes:
      - full_name
      - first_name
    count: 15
    variants:
      - full_name: <fictional full name>
      - first_name: <fictional first name>
  person_2:
    required_attributes:
      - full_name
    count: 15
    variants:
      - full_name: <fictional full name>
places:
  place_1:
    required_attributes:
      - country
      - demonym
    count: 15
    variants:
      - country: <fictional country>
      - demonym: <fictional demonym>
events:
  event_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional event name>
      - type: <event type if needed>
military_orgs:
  military_org_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional military organization name>
enterprise_orgs:
  enterprise_org_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional enterprise organization name>
ngos:
  ngo_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional NGO name>
government_orgs:
  government_org_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional government organization name>
educational_orgs:
  educational_org_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional educational organization name>
media_orgs:
  media_org_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional media organization name>
awards:

```

```

award_1:
  required_attributes:
    - name
    count: 15
  variants:
    - name: <fictional award name>
legals:
  legal_1:
    required_attributes:
      - name
      - reference_code
    count: 15
    variants:
      - name: <fictional legal entity name>
      reference_code: <fictional legal code if needed>
products:
  product_1:
    required_attributes:
      - name
    count: 15
    variants:
      - name: <fictional product name>
...

```

Output rules:

Return YAML only.

Output only the buckets and entity references requested in `required_entities_summary`.

Output only the reference ids listed in `reference_ids_for_this_call`. Any other id is invalid.

Keep only relevant keys in each variant object. Do not write null values.

For organization buckets, write only `{name: ...}` entries inside each variant. The bucket name already carries the taxonomy type.

Set count to the number of distinct valid variants you actually provide for that reference.

Respect `reference_target_counts` and aim to make every requested count equal 15.

Use the rules to make the pool compatible with the later replacement step.

Prefer varied, clean fictional values over tiny spelling variations of the same item.